

Uncertainty-Aware Inference: How Post-Training Quantization Affects LLM Confidence Calibration

Anubha Vyasamudri, Rohit Ramesh, Sanjita Chandan Ballapur, Tamanna Ananna Haque, Vishal Menon

Department of Computer Science

Columbia University

{av3329, rr3713, sb5216, ta2642, vm2820}@columbia.edu

Abstract—Post-training quantization (PTQ) reduces the memory and inference cost of large language models, but its impact on confidence calibration remains insufficiently understood. While quantized models often preserve top-1 accuracy, they can substantially distort output confidence distributions under distribution shift. We present a hardware-grounded study of calibration and serving efficiency across four PTQ methods, GPTQ INT8, GPTQ INT4, AWQ INT4, and NF4 on Mistral-7B. We evaluate calibration using Expected Calibration Error (ECE), Maximum Calibration Error (MCE), Brier Score, entropy, and reliability diagrams across TriviaQA, HellaSwag, and PubMedQA datasets. We additionally profile inference using Pytorch/CUDA-level tracing and Roofline analysis on NVIDIA A100/L4 GPUs. Our results show that we cannot characterize calibration behaviour using a single metric: on PubMedQA, both ECE and Brier Score improve under quantization, while MCE worsens substantially, indicating that average calibration gains can coexist with severe worst-case overconfidence. We further show that kernel implementation dominates serving efficiency much more than bit-width alone. GPTQ INT4 with the Marlin kernel achieves compute-bound inference and $1.78\times$ higher vLLM throughput than FP16, whereas NF4 remains memory-bound despite identical 4-bit precision. Finally, uncertainty-aware routing is beneficial only when the quantized kernel outperforms FP16 under continuous batching. These results demonstrate that evaluating quantized LLMs requires jointly considering calibration, hardware efficiency, and serving dynamics. Code and results are available at <https://github.com/chuanbinp/uncertainty-aware-inference>

Index Terms—Large Language Models, Post-Training Quantization, Confidence Calibration, Expected Calibration Error, Roofline Analysis, CUDA Profiling, Uncertainty Estimation

I. INTRODUCTION

Post-training quantization (PTQ) has become one of the most effective techniques for deploying large language models under practical memory and latency constraints. Quantizing a 7B-parameter model to 4-bit precision can reduce GPU memory usage by approximately 70% while substantially improving serving throughput under batched inference [1]. However, in many high-stakes applications such as medical decision support, financial analysis, and autonomous systems, the reliability of a model’s output confidence is as important as prediction accuracy itself [2]. Quantized models that preserve task accuracy may still become severely miscalibrated, producing overconfident incorrect predictions under distribution shift.

Prior work has shown that quantization can alter confidence distributions and calibration behaviour [3], but existing studies

have largely focused either on model accuracy or quantization efficiency in isolation. A systematic analysis connecting calibration quality, GPU kernel behaviour, and real-world serving throughput across multiple PTQ methods remains limited.

In this work, we study the interaction between quantization, calibration, and hardware efficiency using Mistral-7B across five precision configurations and three benchmark datasets with distinct uncertainty profiles. We evaluate calibration using ECE, MCE, Brier Score, predictive entropy, and reliability diagrams. We additionally profile inference at the CUDA kernel level using NVIDIA Nsight Compute and construct Roofline models from hardware performance counters to analyze arithmetic intensity and memory behaviour across quantization methods.

Our results show that calibration degradation under quantization is highly metric-dependent: configurations that improve ECE under distribution shift can simultaneously exhibit severe worst-case miscalibration captured by MCE. We further demonstrate that kernel implementation, rather than quantization bit-width alone, determines practical serving efficiency. In particular, GPTQ INT4 with the Marlin kernel achieves compute-bound inference and substantially higher vLLM throughput than other quantization schemes with similar memory footprints. Finally, we show that uncertainty-aware routing is viable only when the quantized kernel outperforms FP16 inference under continuous batching.

II. LITERATURE REVIEW

Calibration of Neural Networks: Guo et al. [4] observed that state-of-the-art neural networks tend to be overly confident and devised Expected Calibration Error (ECE), along with temperature scaling, for recalibrating the confidence scores. Naeini et al. [5] formulated Bayesian Binning into Quantiles (BBQ), which is a dynamic counterpart of static binning methods used for calibration estimation. Minderer et al. [6] further highlighted the significant dependence of the calibration phenomenon on architecture choice and data distribution even when there is a distribution shift.

Uncertainty Estimation in LLMs: Kadavath et al. [2] demonstrated that LLMs could self-assess some of their uncertainty, thus paving the way for the utilization of the probabilities assigned to generated outputs. He et al. [7] gave a classification of uncertainty estimations of LLMs based on

token and sequence levels as well as epistemic and aleatoric uncertainties. Semantic entropy was suggested by Kuhn et al. [8], whereas Farquhar et al. [9] extended the approach towards detecting hallucinations in LLMs’ generation processes.

Post-Training Quantization: GPTQ [10] introduced second-order layer-wise post-training quantization for low-bit LLM inference with minimal accuracy degradation. AWQ [11] improved quantization efficiency through activation-aware scaling and fused kernels, while QLoRA [12] introduced the NF4 datatype used in our experiments. The Marlin kernel [13] further enabled efficient fused W4A16-GEMM execution on Ampere and Ada GPUs.

Quantization and Calibration: Proskurina et al. [3] showed that GPTQ quantization can increase calibration error and alter confidence distributions across multiple LLM families. Xia et al. [14] studied the tradeoff between confidence and calibration in quantized networks, and Williams and Aletras [15] demonstrated that calibration degradation under PTQ depends strongly on dataset characteristics and evaluation distribution.

Efficient LLM Serving: vLLM [1] introduced PagedAttention for efficient KV-cache management under continuous batching and has become a standard framework for LLM serving. FlashAttention-2 [16] demonstrated the importance of IO-aware attention kernels to reduce memory traffic, and ATOM [17] proposed serving-aware mixed-precision quantization strategies to balance latency and memory efficiency.

III. METHODOLOGY

A. Models

We evaluate models across three teams covering different architectures and scales, though our work focuses specifically on Team B’s **Mistral-7B-Instruct-v0.2**. The full evaluation suite included:

- **Mistral-7B-Instruct-v0.2 (Team B, primary focus):** A sliding-window attention model with grouped-query attention. We evaluate FP16, GPTQ INT4, GPTQ INT8, AWQ INT4, and NF4. GPTQ weights sourced from TheBloke/Mistral-7B-Instruct-v0.2-GPTQ; AWQ from TheBloke/Mistral-7B-Instruct-v0.2-AWQ.
- **LLaMA-1 7B (Team A, huggyllama/llama-7b):** Full 5-config sweep using TheBloke/LLaMa-7B-GPTQ (with available 4-bit and 8-bit revisions) and TheBloke/LLaMa-7b-AWQ.
- **LLaMA-2 13B (Team C, meta-llama/llama-2-13b-hf):** GPTQ INT4/INT8, AWQ INT4, and NF4 evaluated. FP16 excluded due to VRAM constraints.

B. Hardware

All profiling was conducted on an NVIDIA L4 GPU deployed on Google Cloud Platform on a g2-standard-8 instance (8 vCPUs, 32 GB system memory). The L4 was used for NVIDIA Nsight Systems profiling, as this required the GCP hardware setup and A100 access was unavailable via GCP. Calibration experiments, PyTorch profiling, and vLLM experiments were performed on an NVIDIA A100 (80 GB HBM2e,

2 TB/s DRAM bandwidth, 312 TFLOPS FP16) via Google Colab Pro using PyTorch 2.1.0+cu128.

C. Calibration Evaluation

Calibration is investigated via `lm-evaluation-harness` [18] and `netcal`. Performance metrics include Expected Calibration Error (ECE, 15 bins), Maximum Calibration Error (MCE), Brier score, entropy, and top-1 classification accuracy. ECE is adopted to compute the discrepancy between the confidence and accuracy averages in each bin, whereas MCE records the maximum discrepancy of confidence and accuracy over all bins. Brier score assesses the overall goodness of probabilistic predictions by assigning higher scores to overly confident predictions regardless of their correctness and lower scores for too conservative ones even when they are correct. Entropy serves as an indicator of uncertainty in predictions; that is, larger entropy means lower confidence of output probabilities. Lastly, top-1 classification accuracy measures task performance independent of calibration ability. Visualization of confidence distribution based on quantization is also provided.

D. CUDA Profiling

We investigate inference at both the framework and kernel level.

PyTorch/Kineto Profiling: Every experiment setting is run for 5 autoregressive decode iterations (batch=1) through `torch.profiler` with CUDA profiling on. Throughput, peak GPU memory usage, duration, and prominent CUDA kernels are reported.

Nsight Compute Profiling: To analyze Roofline, individual forward runs (prefill, seq=128) were investigated via NVIDIA’s Nsight Compute (`ncu`) to gather metrics for SM efficiency, DRAM bandwidth, and tensor operation counts. Arithmetic intensity was estimated through measured floating point operations and memory bandwidth.

E. Serving Benchmarks and Routing

Serving throughput is measured using continuous batching, with 10 concurrent prompts being issued to the server. As support for NF4 is not yet available within vLLM [1], we use the HuggingFace batched inference implementation as a performance lower bound for comparison purposes.

Furthermore, we experiment with the idea of uncertainty-aware routing, where any prompt that is processed with confidence greater than τ is handled by the quantized model, while any other is promoted to FP16 execution.

IV. EXPERIMENTAL RESULTS

A. Calibration Results

Table I presents calibration and accuracy results for Mistral-7B. Accuracy is remarkably stable across quantization methods on TriviaQA and HellaSwag, with ECE degradation of only +0.002 to +0.015 and minimal Brier score rise (at most +0.010 on TriviaQA). Average entropy increases slightly across all quantized configurations, consistent with mild confidence diffusion from weight rounding. PubMedQA accuracy is

TABLE I: Mistral-7B Full Calibration Metrics Across Quantization Configurations

Dataset	Config	ECE	MCE	Brier	Avg. Entropy	Acc	Δ ECE	Δ Acc
TriviaQA	FP16	0.143	0.457	0.222	0.402	70.1%	-	-
	GPTQ INT4	0.152	0.194	0.231	0.431	68.1%	+0.009	-2.0%
	GPTQ INT8	0.155	0.312	0.235	0.428	67.8%	+0.012	-2.3%
	AWQ INT4	0.155	0.340	0.229	0.415	68.5%	+0.012	-1.6%
	NF4	0.158	0.315	0.230	0.397	68.7%	+0.015	-1.4%
HellaSwag	FP16	0.277	0.348	0.197	1.088	82.9%	-	-
	GPTQ INT4	0.289	0.487	0.207	1.116	82.5%	+0.012	-0.4%
	GPTQ INT8	0.283	0.353	0.206	1.114	81.9%	+0.006	-1.0%
	AWQ INT4	0.282	0.352	0.205	1.114	81.9%	+0.005	-1.0%
	NF4	0.279	0.355	0.201	1.097	82.6%	+0.002	-0.3%
PubMedQA	FP16	0.229	0.330	0.302	0.656	49.1%	-	-
	GPTQ INT4	0.077	0.302	0.252	0.938	49.3%	-0.152	+0.2%
	GPTQ INT8	0.103	0.944	0.251	0.924	46.1%	-0.126	-3.0%
	AWQ INT4	0.067	0.939	0.250	0.934	49.2%	-0.162	+0.1%
	NF4	0.236	0.330	0.304	0.663	48.1%	+0.007	-1.0%

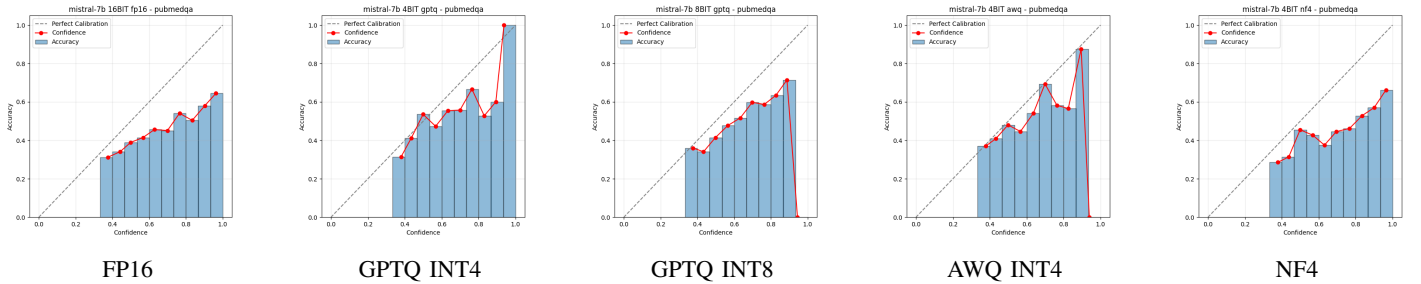


Fig. 1: Reliability diagrams for Mistral-7B on PubMedQA (OOD).

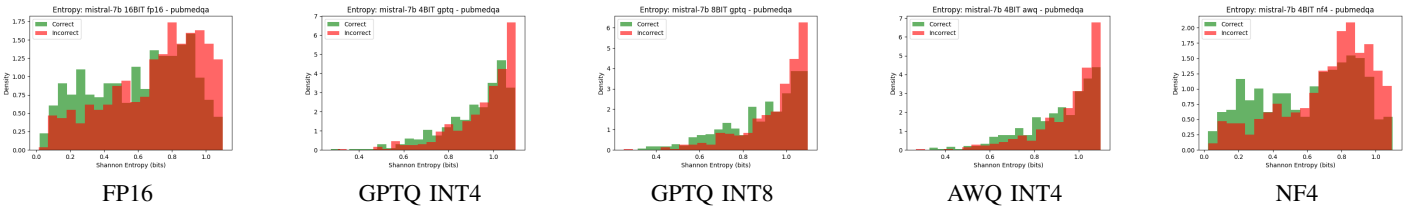


Fig. 2: Predictive entropy distributions for Mistral-7B on PubMedQA.

uniformly low (46–49%) across all configurations, confirming the OOD nature of the task.

On PubMedQA, quantization produces a split calibration profile that exposes the limits of any single metric (Figs. 1, 2). GPTQ INT4, GPTQ INT8, and AWQ INT4 all reduce ECE (from 0.229 to as low as 0.067) and improve Brier Score by 0.050–0.052, indicating genuine average calibration improvement. FP16 is so severely overconfident on this OOD task that quantization-induced confidence reduction is beneficial on average. This result is corroborated by average entropy nearly doubling from FP16 (0.656) to GPTQ INT4 (0.938),

confirming that confidence mass is genuinely spreading rather than merely redistributing. MCE, however, tells a different story: GPTQ INT8 MCE rises from 0.330 to 0.944 and AWQ INT4 to 0.939, revealing that while average calibration improves, confidence mass concentrates solely in a single bin. This divergence, ECE and Brier improving while MCE worsens is diagnostic of a failure mode where quantization redistributes overconfidence into a concentrated worst-case bin rather than eliminating it, a pattern invisible to any single metric alone.

NF4 is the only quantized method that does not exhibit

this pattern. Its ECE, MCE, Brier Score, and entropy all remain close to FP16 under OOD shift. Unlike GPTQ and AWQ, which map weights to uniform integer grids, NF4 places more quantization levels where pretrained weights are dense, minimizing structured perturbation to the weight distribution. This carries through to the output confidence distribution, suggesting that distribution-aware quantization preserves calibration where uniform methods fail.

Table II summarizes FP16 baseline calibration across all three models. The most consistent finding is that task structure dominates calibration: every model follows the same ECE ordering: TriviaQA (best) $\rightarrow$ PubMedQA $\rightarrow$ HellaSwag(worst) regardless of model family or size. The HellaSwag dataset was constructed such that wrong answers are as likely as correct ones; given the language-modeling objective of predicting the most likely next token, this construct actively games the model’s confidence signal and drives overconfidence. TriviaQA, by contrast, requires direct factual recall without candidate comparison, producing more honest uncertainty estimates.

TABLE II: FP16 Baseline Calibration - Cross-Model Comparison

Model	Dataset	ECE	Accuracy
Mistral-7B	TriviaQA	0.143	70.1%
	HellaSwag	0.277	82.9%
	PubMedQA	0.229	49.1%
LLaMA-1 7B	TriviaQA	0.084	68.5%
	HellaSwag	0.309	73.9%
	PubMedQA	0.239	53.6%
LLaMA-2 13B	TriviaQA	0.167	76.7%
	HellaSwag	0.335	77.5%
	PubMedQA	0.229	47.0%

B. GPU-Level Profiling Results

Table III presents decode-mode profiling results for Mistral-7B under single-query inference on the L4. The memory reduction from quantization is substantial: GPTQ INT4 and AWQ INT4 both use approximately 4.2–4.3 GB peak GPU memory, a 70% reduction from FP16’s 14.5 GB. Despite this footprint advantage, all quantized methods show *lower* single-query decode throughput than FP16 (13.1 tok/s), because at batch=1, inference is memory bandwidth-bound and the overhead of dequantization consumes the bandwidth savings from smaller weights.

Arithmetic intensity (AI) reveals sharp differences in how each method uses the GPU. FP16 achieves AI = 57.7 FLOPs/byte using standard `s16816gemv` cuBLAS kernels, firmly memory-bound on both the A100 and L4. GPTQ INT4 raises AI to 195.1 FLOPs/byte, a 3.4 $\times$ increase, because the Marlin-style dequantization kernel packs more arithmetic per memory transaction, though at the cost of higher avg CUDA time per step (13,737 ms vs 7,773 ms for FP16),

reflecting the dequantization overhead paid in kernel time even when not paid in memory bandwidth.

AWQ INT4 shows a striking anomaly: AI = 3.2 FLOPs/byte, *lower* than FP16, despite nominally 4-bit weights. This result likely occurs because the `awq_gemm_kernel` operates as a fused grouped GEMV (matrix-vector product) rather than a full GEMM, and the profiler attributes only 13.2 BFLOPs per step to AWQ, opposed to the 836.8 BFLOPs for FP16 and GPTQ. The AWQ kernel trades raw FLOPs for fused memory access, which is efficient for single-batch prefill but explains the low vLLM throughput under batched serving (IV-D).

NF4 achieves AI = 36.1 FLOPs/byte, lower than FP16, because the `kgemv_4bit_inference_naive` kernel is genuinely naive: it lacks the memory coalescing and tensor-core scheduling of cuBLAS or Marlin. The high `aten::copy_` and `aten::to` call counts (29,645 and 44,196 respectively) further indicate repeated dtype cast overhead from dequantization-on-the-fly. NF4 also shows the highest step-to-step timing variance (CV = 1.5% vs 0.5–0.8% for other configs), consistent with irregular memory access patterns.

C. Roofline Analysis

Figure 3 and Table IV summarize the hardware-counter Roofline analysis for Mistral-7B prefill (seq=128) on the NVIDIA L4.

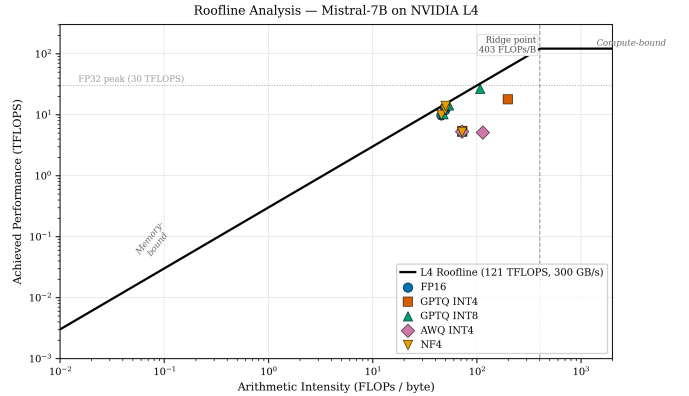


Fig. 3: Roofline analysis for Mistral-7B prefill (seq=128) on NVIDIA L4. Kernel implementation determines hardware efficiency more strongly than quantization bit-width alone.

The dominant observation is that kernel implementation impacts hardware efficiency more strongly than quantization bit-width alone. GPTQ INT4 with the Marlin kernel achieves the highest effective compute utilization among all configurations, operating at AI=951 FLOPs/byte with 82% SM utilization, making it the only configuration that is clearly compute-bound. AWQ achieves even higher arithmetic intensity (AI=1865 FLOPs/byte) through its fused GEMV kernel, though with lower SM utilization (67%). In contrast, NF4 remains near the roofline ridge point: the `kDequantizeBlockwise` kernel operates at AI=388 FLOPs/byte with both high SM and memory utilization, while the subsequent FP16 GEMM remains memory-bound due to the absence of a fused 4-bit GEMM imple-

TABLE III: Mistral-7B Single-Query Inference Profile (A100-80GB, batch=1, 5 decode steps)

Config	Top Kernel	Tok/s	Peak Mem (GB)	AI (FLOPs/B)	Avg CUDA (ms)	Top Kernel (%)
FP16	<i>s16816gemm</i>	13.1	14.5	57.7	7,773	8.2%
GPTQ INT4	<i>QuantLinearFn</i>	8.0	4.3	195.1	13,737	12.7%
GPTQ INT8	<i>QuantLinearFn</i>	7.9	7.8	107.2	13,731	12.6%
AWQ INT4	<i>WQLinearMMFn</i>	9.1	4.2	3.2	9,093	9.9%
NF4	<i>gemv_4bit</i>	6.6	4.4	36.1	11,171	3.8%

Profiled on A100-80GB (Ampere). AI computed from Kineto FLOP counts and DRAM throughput.

TABLE IV: Hardware-Counter Roofline: Dominant Kernel per Config (Mistral-7B, prefill, seq=128)

Config	Kernel	SM%	MEM%	AI (FLOPs/B)
FP16	s16816gemm 256×128	52	87	243
GPTQ INT4	Marlin GEMM (W4A16)	82	35	951
GPTQ INT8	s16816gemm 128×128	58	88	266
AWQ INT4	awq_gemm_kernel	67	14	1865
NF4	kDequantizeBlockwise	81	84	388

mentation. These results show that we determine arithmetic intensity and utilization primarily using kernel design rather than quantization precision alone.

D. vLLM Serving Throughput

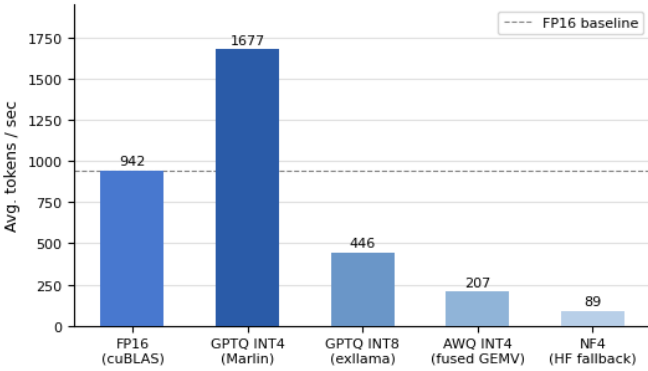


Fig. 4: vLLM batched serving throughput for Mistral-7B quantization configurations (batch=10, NVIDIA L4).

TABLE V: Mistral-7B vLLM Serving Throughput (batch=10, n_output=128, NVIDIA L4)

Config	Kernel	Avg Tok/s	vs FP16	Peak Mem (GB)
FP16	cuBLAS	942.1	1.00×	71.6
GPTQ INT4	gptq_marlin	1677.1	1.78×	71.7
GPTQ INT8	gptq (exllama)	445.5	0.47×	72.1
AWQ INT4	fused_GEMV (AWQ)	207.2	0.22×	72.1
NF4 [†]	bitsandbytes (HF)	89.2	0.09×	4.9

[†]vLLM does not support bitsandbytes NF4; HF batched fallback used as lower bound.

Table V and Fig. 4 present throughput measurements under vLLM continuous batching (batch=10). GPTQ INT4 with the Marlin kernel achieves the highest serving throughput at $1.78\times$ FP16 throughput, consistent with its compute-bound Roofline position. In contrast, GPTQ INT8 with the exllama kernel achieves only $0.47\times$ FP16 throughput, demonstrating that

serving efficiency depends more strongly on kernel implementation than quantization bit-width.

AWQ INT4 performs particularly poorly under vLLM serving despite its high arithmetic intensity during prefill. This result occurs because optimization of the fused AWQ GEMV kernel was for single-request prefill rather than large-batch decode workloads under continuous batching. NF4 achieves the lowest throughput and requires a HuggingFace fallback implementation because bitsandbytes NF4 is unsupported in vLLM. Its reported peak memory is therefore not directly comparable to vLLM configurations, as the HuggingFace implementation does not allocate the unified KV-cache memory pool used by vLLM.

E. Uncertainty-Based Routing

Table VI summarizes routing viability across model and quantization configurations on PubMedQA. Only when the performance gain from the quantized kernel is higher than FP16 on vLLM continuous batching does routing become favorable.

TABLE VI: Optimal routing thresholds on PubMedQA. Positive savings indicate the quantized configuration outperforms FP16 under vLLM serving.

Model	Config	τ^*	Savings	Viable?
LLaMA-1 7B	GPTQ INT4	0.00	+69.8%	✓
LLaMA-1 7B	GPTQ INT8	0.35	+67.8%	✓
LLaMA-2 13B	GPTQ INT4	0.00	+20.2%	✓
Mistral-7B	GPTQ INT4	0.42	−108.8%	×
Mistral-7B	GPTQ INT8	0.55	−59.4%	×
Mistral-7B	AWQ INT4	0.39	−80.0%	×
Mistral-7B	NF4	0.63	−122.8%	×

Viability of routing is largely defined by kernel performance rather than the quantization precision or quality of calibration. LLaMA-1 7B GPTQ INT4 demonstrates the best results in terms of routing due to its vLLM throughput on Marlin. Most Mistral-7B quantizations are slower than FP16 and thus result in negative cost savings.

F. Knowledge Distillation Recovery

We analyze the soft-label knowledge distillation from an FP16 Mistral-7B teacher to an NF4 student at several distillation temperatures. Lower temperatures show partial improvements in ECE with moderate accuracy drops, but higher

temperatures cause severe accuracy collapse even though ECE seems to have improved.

Distillation at high temperatures consistently raises MCE, showing that the probability mass gets excessively flattened rather than being properly calibrated. This matches the failure point seen during quantization, where the calibration measures on average get better, but the worst-case measures worsen. Among the evaluated settings, $T = 2$ provides the best calibration–accuracy tradeoff without triggering catastrophic MCE growth.

As shown in 5 and Figures 7 (included in the Appendix), knowledge distillation results reveal two consistent patterns. Firstly, symmetric temperature scaling shows that lower temperatures ($T = 1$, $T = 2$) yield the best calibration–accuracy tradeoff, with performance degrading monotonically as temperature increases. Secondly, asymmetric dual-temperature configurations generally fail to recover the accuracy of the symmetric $T = 1$ baseline. While ECE improvements are observed on HellaSwag, these come at the cost of accuracy, suggesting a calibration–accuracy tradeoff introduced by asymmetric softening.

G. Pareto Analysis

Fig. 6 (included in the Appendix) shows the Pareto frontiers across calibration, accuracy, and throughput for all model-configuration pairs. No configuration simultaneously optimizes all three objectives. GPTQ INT4 configurations dominate throughput-oriented deployment due to Marlin-backed serving efficiency, while AWQ INT4 achieves the best average calibration. The calibration frontier is comparatively narrow relative to throughput, suggesting that kernel implementation has a substantially larger effect on serving efficiency than on average calibration quality.

V. DISCUSSION

In-Distribution and OOD Calibration Behave Differently Under Quantization. In in-distribution tasks, quantization causes only modest ECE degradation with minimal accuracy loss. Under OOD shift (PubMedQA), the picture is more complex: ECE and Brier Score improve while MCE worsens sharply, indicating that average calibration gains can coexist with worst-case overconfidence concentrated in a single bin. NF4 is the only configuration that consistently preserves the FP16 calibration profile under distribution shift, suggesting that distribution-aware quantization better maintains output confidence structure. These results demonstrate that ECE alone is insufficient for evaluating calibration robustness under OOD shift.

Kernel Design Dominates Inference Efficiency. The $19\times$ arithmetic intensity gap between NF4 and AWQ at identical 4-bit precision shows that serving efficiency is determined more by kernel implementation than by bit-width itself. Marlin-backed GPTQ INT4 achieves compute-bound inference and the highest vLLM throughput, whereas NF4 remains memory-bound due to explicit dequantization overheads.

Arithmetic Intensity Does Not Predict Serving Throughput. Although AWQ achieves the highest arithmetic intensity during prefill, it underperforms significantly under vLLM continuous batching. This discrepancy arises because the AWQ fused GEMV kernel is optimized for single-request prefill rather than batched decode workloads. Practical serving performance therefore depends not only on hardware efficiency metrics, but also on compatibility with the serving stack and batching regime.

Routing Viability Is Model and Kernel-Dependent Uncertainty-based routing is beneficial only when the quantized kernel is faster than FP16 under vLLM serving. Across our experiments, routing viability was determined primarily by kernel throughput characteristics rather than quantization bit-width or confidence threshold selection.

Quantization and Distillation Share a Common Failure Mode. Knowledge distillation at high temperatures reproduces the same MCE failure pattern: ECE and Brier Score improve while MCE increases sharply, indicating probability mass flattening rather than proper redistribution. $T = 2$ provides the best calibration–accuracy tradeoff, and MCE should be treated as a necessary diagnostic in any post-hoc model modification pipeline.

VI. CONCLUSION

This work presents a hardware-grounded study of how post-training quantization affects confidence calibration and serving efficiency in large language models. We show that calibration under quantization cannot be characterized solely by accuracy or ECE. Under a distribution shift, quantized models may exhibit improved average calibration while simultaneously developing severe worst-case overconfidence captured by MCE. We further demonstrate that practical inference efficiency depends more on kernel implementation than on quantization bit-width itself, with Marlin-backed GPTQ INT4 achieving compute-bound execution and substantially higher vLLM throughput, whereas NF4 remains memory-bound despite identical 4-bit precision. These hardware-level differences directly determine the viability of uncertainty-aware serving strategies, as routing is beneficial only when the quantized kernel outperforms FP16 inference under continuous batching. Overall, our results highlight the need to jointly evaluate calibration robustness, kernel-level execution behaviour, and serving efficiency when deploying quantized LLMs.

VII. AI USE DISCLOSURE

AI-assisted tools were used in a limited capacity for debugging environment and CUDA dependency issues, resolving package conflicts, and proofreading portions of the final report. All experiments, profiling runs, numerical results, analyses, and interpretations were independently conducted and verified by the authors using raw outputs from calibration, vLLM, PyTorch profiler, and Nsight Compute traces. No AI-generated content was used for experimental findings or technical conclusions.

REFERENCES

- [1] W. Kwon, Z. Li, S. Zhuang *et al.*, “Efficient memory management for large language model serving with PagedAttention,” in *Proceedings of the ACM Symposium on Operating Systems Principles (SOSP)*, 2023.
- [2] S. Kadavath, T. Conerly, A. Askell *et al.*, “Language models (mostly) know what they know,” *arXiv preprint arXiv:2207.05221*, 2022.
- [3] I. Proskurina *et al.*, “When quantization affects confidence of large language models?” in *Findings of the Association for Computational Linguistics: NAACL*, 2024.
- [4] C. Guo, G. Pleiss, Y. Sun, and K. Q. Weinberger, “On calibration of modern neural networks,” in *Proceedings of the 34th International Conference on Machine Learning (ICML)*, 2017.
- [5] M. P. Naeini, G. F. Cooper, and M. Hauskrecht, “Obtaining well calibrated probabilities using Bayesian binning into quantiles,” in *Proceedings of the 29th AAAI Conference on Artificial Intelligence*, 2015.
- [6] M. Minderer, J. Djolonga, R. Romijnders, F. Hubis, X. Zhai, N. Houlsby, D. Tran, and M. Lucic, “Revisiting the calibration of modern neural networks,” in *Advances in Neural Information Processing Systems (NeurIPS)*, 2021.
- [7] J. He *et al.*, “Survey of uncertainty estimation in large language models: Sources, methods, applications, and challenges,” *arXiv preprint*, 2025.
- [8] L. Kuhn, Y. Gal, and S. Farquhar, “Semantic uncertainty: Linguistic invariances for uncertainty estimation in natural language generation,” in *International Conference on Learning Representations (ICLR)*, 2023.
- [9] S. Farquhar, J. Kossen, L. Kuhn, and Y. Gal, “Detecting hallucinations in large language models using semantic entropy,” *Nature*, 2024.
- [10] E. Frantar, S. Ashkboos, T. Hoefler, and D. Alistarh, “GPTQ: Accurate post-training quantization for generative pre-trained transformers,” in *International Conference on Learning Representations (ICLR)*, 2023.
- [11] J. Lin, J. Tang, H. Tang, S. Yang, X. Dang, C. Gan, and S. Han, “AWQ: Activation-aware weight quantization for LLM compression and acceleration,” in *Proceedings of Machine Learning and Systems (MLSys)*, 2024.
- [12] T. Dettmers, A. Pagnoni, A. Holtzman, and L. Zettlemoyer, “QLoRA: Efficient finetuning of quantized LLMs,” in *Advances in Neural Information Processing Systems (NeurIPS)*, 2023.
- [13] E. Frantar and D. Alistarh, “MARLIN: Mixed-precision autoregressive parallel inference on large language models,” *arXiv preprint arXiv:2408.11743*, 2024.
- [14] Z. Xia *et al.*, “An underexplored dilemma between confidence and calibration in quantized neural networks,” *arXiv preprint arXiv:2106.XXXXX*, 2021.
- [15] J. Williams and N. Aletras, “On the impact of calibration data in post-training quantization and pruning,” in *Proceedings of the 62nd Annual Meeting of the Association for Computational Linguistics (ACL)*, 2024.
- [16] T. Dao, “FlashAttention-2: Faster attention with better parallelism and work partitioning,” in *International Conference on Learning Representations (ICLR)*, 2024.
- [17] Y. Zhao *et al.*, “ATOM: Low-bit quantization for efficient and accurate LLM serving,” in *Proceedings of Machine Learning and Systems (MLSys)*, 2024.
- [18] L. Gao *et al.*, “A framework for few-shot language model evaluation,” EleutherAI lm-evaluation-harness, 2021. [Online]. Available: <https://github.com/EleutherAI/lm-evaluation-harness>

APPENDIX

This appendix contains supplementary calibration and knowledge distillation figures, experiments conducted by complementary teams that are referenced in the main paper but omitted from the primary discussion.

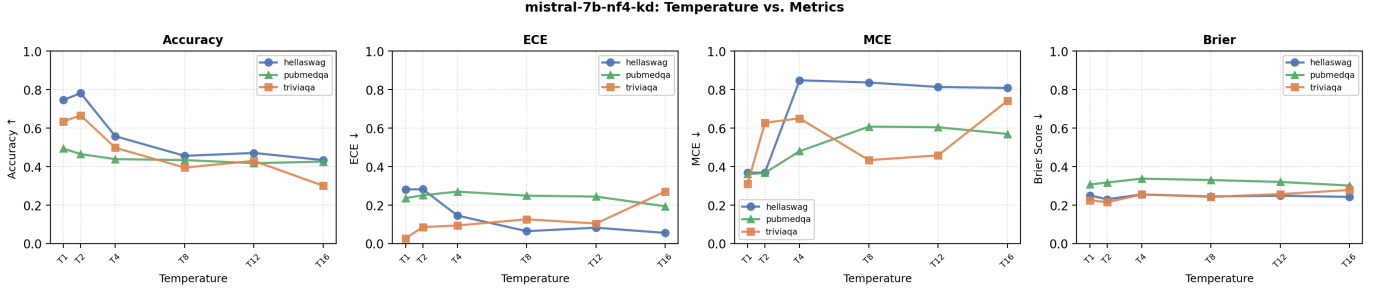
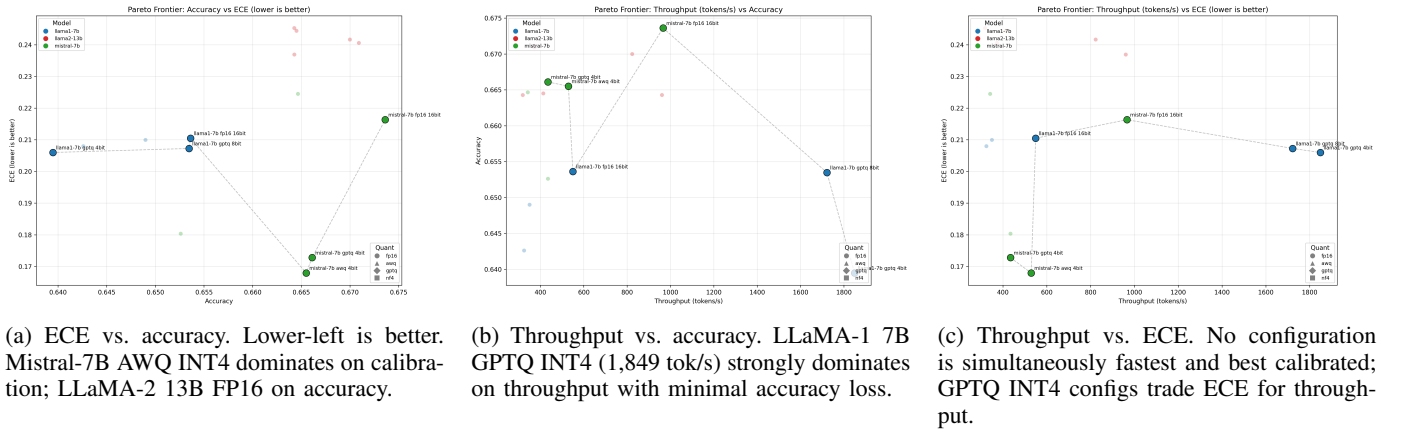


Fig. 5: Knowledge Distillation Temperature vs Calibration Metrics. Lower temperatures (T1–T2) yield higher accuracy and better calibration across general-domain datasets. Performance degrades monotonically with increasing temperature, with T16 showing the largest drop.



(a) ECE vs. accuracy. Lower-left is better. Mistral-7B AWQ INT4 dominates on calibration; LLaMA-2 13B FP16 on accuracy. (b) Throughput vs. accuracy. LLaMA-1 7B GPTQ INT4 (1,849 tok/s) strongly dominates on throughput with minimal accuracy loss. (c) Throughput vs. ECE. No configuration is simultaneously fastest and best calibrated; GPTQ INT4 configs trade ECE for throughput.

Fig. 6: Two-dimensional Pareto frontiers across all 15 model-configuration pairs (3 models $\times$ 5 configs), averaged across TriviaQA, HellaSwag, and PubMedQA. Pareto-dominant points are connected; dominated points shown in grey.

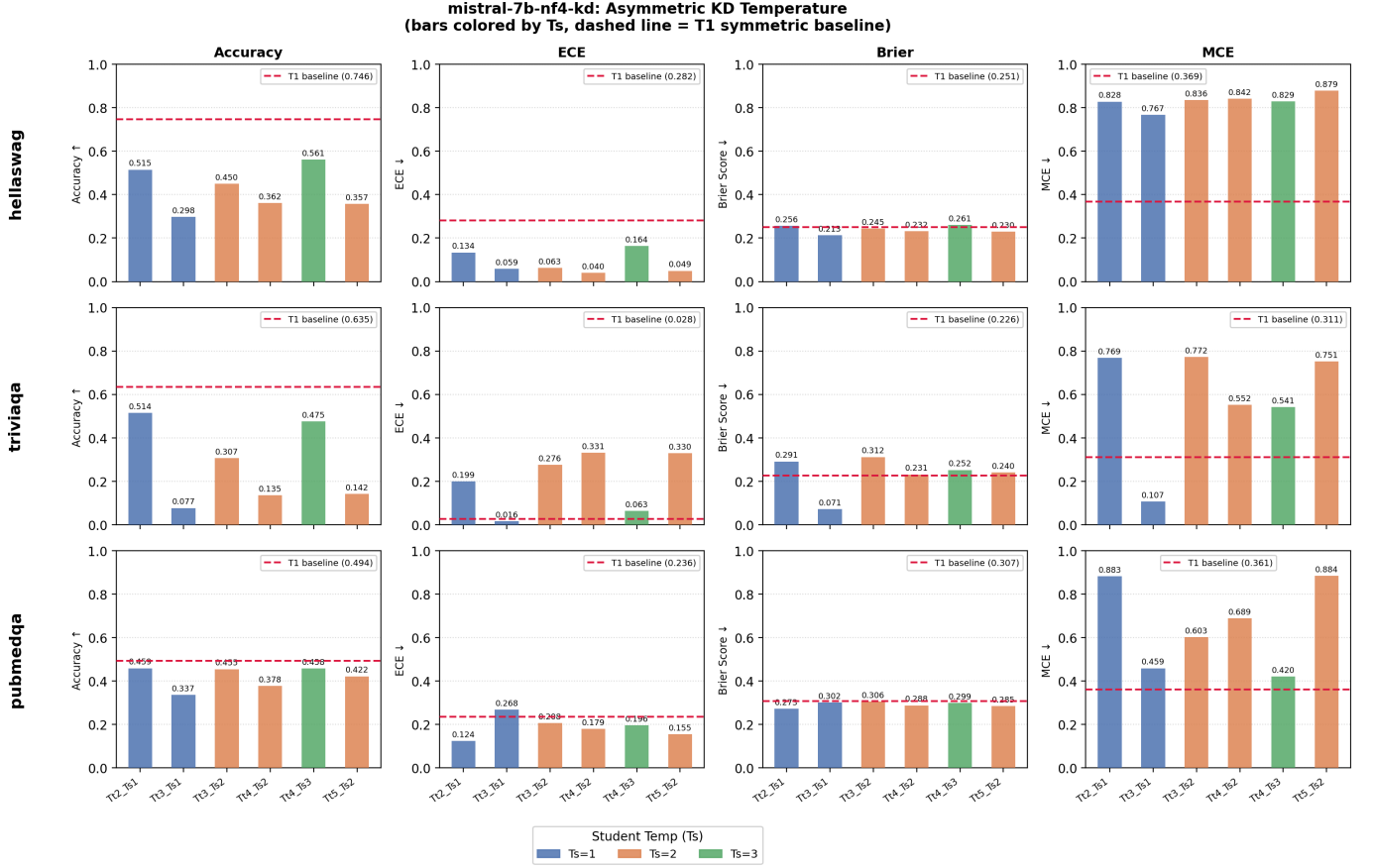


Fig. 7: Asymmetric Knowledge Distillation Temperature Scaling on Mistral-7B NF4-KD. Each bar represents a dual-temperature configuration (T_t , T_s), where T_t is the teacher temperature and T_s is the student temperature; bars are colored by T_s . The dashed red line indicates the symmetric T_1 baseline.